

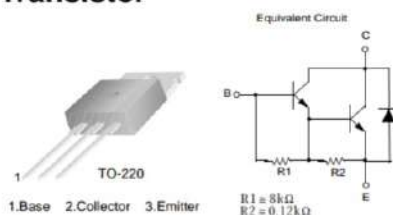


ON Semiconductor®

TIP120 / TIP121 / TIP122 NPN Epitaxial Darlington Transistor

Features

- Medium Power Linear Switching Applications
- Complementary to TIP125 / TIP126 / TIP127



Ordering Information

Part Number	Top Mark	Package	Packing Method
TIP120	TIP120	TO-220 3L (Single Gauge)	Bulk
TIP120TU	TIP120	TO-220 3L (Single Gauge)	Rail
TIP121	TIP121	TO-220 3L (Single Gauge)	Bulk
TIP121TU	TIP121	TO-220 3L (Single Gauge)	Rail
TIP122	TIP122	TO-220 3L (Single Gauge)	Bulk
TIP122TU	TIP122	TO-220 3L (Single Gauge)	Rail

Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only. Values are at $T_C = 25^\circ\text{C}$ unless otherwise noted.

Symbol	Parameter	Value	Unit
V_{CB0}	Collector-Base Voltage	TIP120	60
		TIP121	80
		TIP122	100
V_{CE0}	Collector-Emitter Voltage	TIP120	60
		TIP121	80
		TIP122	100
V_{EB0}	Emitter-Base Voltage	5	V
I_C	Collector Current (DC)	5	A
I_{CP}	Collector Current (Pulse)	8	A
I_B	Base Current (DC)	120	mA
T_J	Junction Temperature	150	$^\circ\text{C}$
T_{STG}	Storage Temperature Range	-65 to 150	$^\circ\text{C}$

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Publication Order Number:
TIP122/D

Thermal Characteristics

Values are at $T_C = 25^\circ\text{C}$ unless otherwise noted.

Symbol	Parameter	Value	Unit
P_C	Collector Dissipation ($T_A = 25^\circ\text{C}$)	2	W
	Collector Dissipation ($T_C = 25^\circ\text{C}$)	65	

Electrical Characteristics

Values are at $T_C = 25^\circ\text{C}$ unless otherwise noted.

Symbol	Parameter	Conditions	Min.	Max.	Unit
$V_{CE0(sus)}$	Collector-Emitter Sustaining Voltage	TIP120 $I_C = 100\text{ mA}, I_B = 0$	60		V
		TIP121	80		
		TIP122	100		
I_{CEO}	Collector Cut-Off Current	TIP120 $V_{CE} = 30\text{ V}, I_B = 0$		0.5	mA
		TIP121 $V_{CE} = 40\text{ V}, I_B = 0$		0.5	
		TIP122 $V_{CE} = 50\text{ V}, I_B = 0$		0.5	
I_{CBO}	Collector Cut-Off Current	TIP120 $V_{CB} = 60\text{ V}, I_E = 0$		0.2	mA
		TIP121 $V_{CB} = 80\text{ V}, I_E = 0$		0.2	
		TIP122 $V_{CB} = 100\text{ V}, I_E = 0$		0.2	
I_{EBO}	Emitter Cut-Off Current	$V_{EB} = 5\text{ V}, I_C = 0$		2	mA
h_{FE}	DC Current Gain ⁽¹⁾	$V_{CE} = 3\text{ V}, I_C = 0.5\text{ A}$	1000		
		$V_{CE} = 3\text{ V}, I_C = 3\text{ A}$	1000		
$V_{CE(sat)}$	Collector-Emitter Saturation Voltage ⁽¹⁾	$I_C = 3\text{ A}, I_B = 12\text{ mA}$		2.0	V
		$I_C = 5\text{ A}, I_B = 20\text{ mA}$		4.0	
$V_{BE(on)}$	Base-Emitter On Voltage ⁽¹⁾	$V_{CE} = 3\text{ V}, I_C = 3\text{ A}$		2.5	V
C_{ob}	Output Capacitance	$V_{CB} = 10\text{ V}, I_E = 0,$ $f = 0.1\text{ MHz}$		200	pF

Note:

1. Pulse test: $p_w \leq 300\text{ }\mu\text{s}$, duty cycle $\leq 2\%$.

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